



The Scots College

Year 12 Mathematics Extension 2

Pre-Trial

April 2010

Name: _____

General Instructions

- Working time - 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Standard Integrals Table attached

TOTAL MARKS: 80

THERE ARE 4 QUESTIONS, EACH WORTH 20 MARKS.

WEIGHTING: 30 %

Question 1: (Marks 20)

a)

Let $z = \cos \theta + i \sin \theta$, show that

2

i) $z^n + \frac{1}{z^n} = 2 \cos n\theta$,

ii) $z^n - \frac{1}{z^n} = 2i \sin n\theta$

2

b)

For each of the following, find and sketch the locus of the point representing the complex number Z :

i) $\operatorname{Re}(iz) = -1$

2

ii) $|z + 1 + i| = 2$

2

iii) $\left| \frac{z-1}{z+1} \right| \leq 2$

2

iv) $\operatorname{Arg}(z - i) = \frac{\pi}{4}$

2

Continue.....

Question 1 continued.....

c)

- i) P is a point in an Argand diagram representing a non-zero complex number z such that $\arg\left(\frac{z}{1-i}\right) = \frac{\pi}{2}$

a) Find $\arg(1-i)$ and $\arg z$. 2

b) Hence or otherwise sketch the locus of P in an Argand diagram. 1

- ii) R is a point on the Argand diagram representing a non-zero complex number w , such that 2

$$w^3 = (\bar{w})^3 \text{ and } \frac{\pi}{2} < \arg w < \pi$$

Find $\arg w$

(Hint: you may let $w = r(\cos \theta + i \sin \theta)$

- i) It is given that in part (i), $|z| = 2\sqrt{2}$ and in part (ii), $|w| = 2$, furthermore, $OPQR$ is a parallelogram in an Argand diagram, where O represents the complex number 0. 3

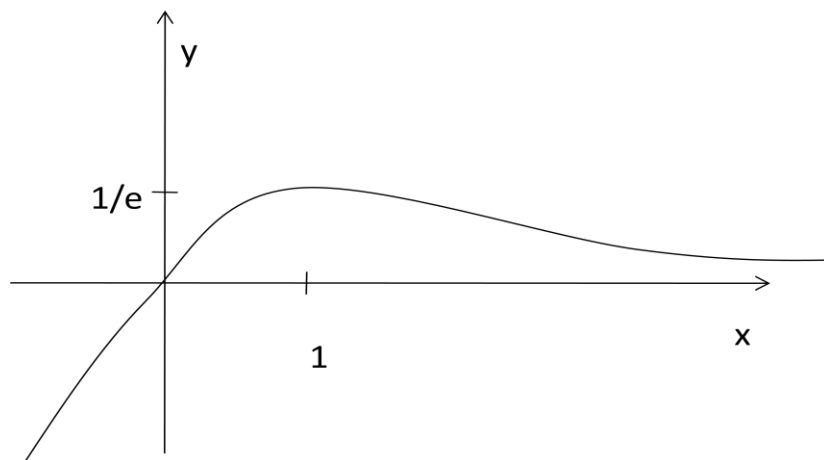
Find the complex number represented by the point Q , giving your answer in standard form.

Continue.....

Question 2: (Marks 20)

a)

The graph of $y = x e^{-x}$ is shown below :



On separate axes, sketch the following curves. Indicate clearly any turning points, asymptotes and intercepts with the coordinate axes.

- | | | |
|------|-----------------------------|---|
| i) | $y = x^2 e^{-2x}$ | 2 |
| ii) | $y = \log_e(x e^{-x})$ | 2 |
| iii) | $y = \frac{1}{x^2 e^{-2x}}$ | 2 |
| iv) | $y = e^{x e^{-x}}$ | 2 |

b)

Draw a sketch of the following, showing important features :

- | | | |
|-----|--------------------------|---|
| i) | $f(x) = \frac{4}{x} - x$ | 2 |
| ii) | $y = \sqrt{f(x)}$ | 2 |

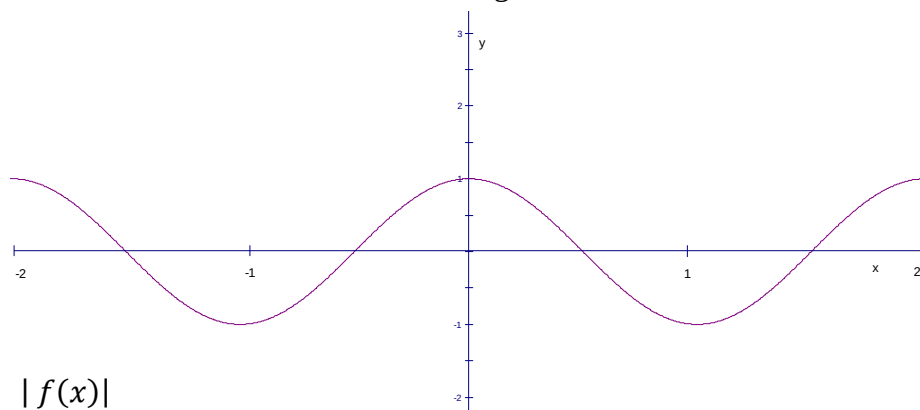
Continue.....

Question 2 continued.....

c)

The diagram below shows the graph of the continuous function $y = f(x)$. Critical points occur at $x = -2, -1, 0, 1, 2$

Draw separate sketches for each of the following:



i) $|y| = |f(x)|$

ii) $y = \frac{1}{f(x)}$

2

2

iii) $y = \sqrt{f(x)}$

2

iv) $y = xf(x)$

2

Question 3: (Marks 20)

a)

The hyperbola H has equation $xy = 16$

- i) Sketch this hyperbola and indicate on your diagram the positions and co-ordinates of all points at which the curve intersects the axes of symmetry.

2

- ii) $P\left(4p, \frac{4}{p}\right)$ and $Q\left(4q, \frac{4}{q}\right)$ are two distinct arbitrary points on H ($p > 0$, $q > 0$). Find the equation of the chord PQ .

2

- iii) Prove that the equation of the tangent at P is $x + p^2y = 8p$.

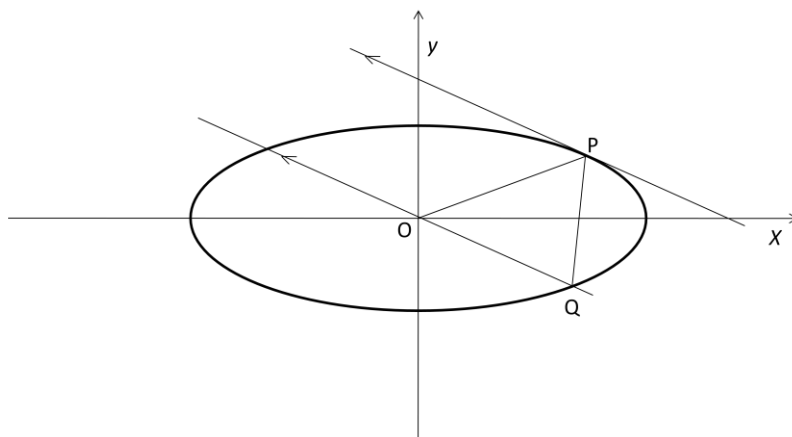
2

- iv) The tangents at P and Q intersect at T . Find the coordinates of T .

2

Question 3 continued.....

b)



In the diagram above $P(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where P lies in the first quadrant.

A straight line through the origin parallel to the tangent at P meets the ellipse at the point Q , where P and Q both lie on the same side of the y -axis.

- i) Prove that the equation of the line OQ is $x b \cos \theta + y a \sin \theta = 0$. 2
- ii) Find the coordinates of the point Q given that Q lies in the fourth quadrant. 3
- iii) Prove that the area of the ΔOPQ is independent of the position of P . 3

c)

- i) Express $\frac{x^2 - 2x - 3}{(x + 2)(x^2 + 1)}$ as $\frac{A}{x + 2} + \frac{Bx + C}{x^2 + 1}$ 2

- ii) Hence evaluate $\int_0^1 \frac{x^2 - 2x - 3}{(x + 2)(x^2 + 1)} dx$. 2

Continue.....

Question 4: (Marks 20)

a)

$P(x)$ is a cubic polynomial with real coefficients . One zero of $P(x)$ is $1 + 2i$, the constant term is -15 and $P(2) = 5$.

5

Write $P(x)$ with real coefficients.

b)

The equation $x^3 - 4x + 5 = 0$ has roots α , β and γ .

i) Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

2

ii) Find the equation whose roots are $\alpha + \beta$, $\beta + \gamma$, $\gamma + \alpha$.

2

c)

Factorise $x^4 - 2x^2 - 15$ over the Complex number field.

2

d)

Solve the equation $x^4 - 5x^3 + 4x^2 + 3x + 9$ over complex field , given it has a root of multiplicity 2.

3

e)

If the roots of $x^3 - px + q = 0$ are α , β and γ , find in terms of p and q a cubic equation with roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$.

2

f)

i) By considering the expansion of $(\cos \theta + i \sin \theta)^5$ and by using De Moivre's theorem, show that

2

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

ii) Hence find all the four roots of the equation $16x^4 - 20x^2 + 5 = 0$.

2

END OF EXAM

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

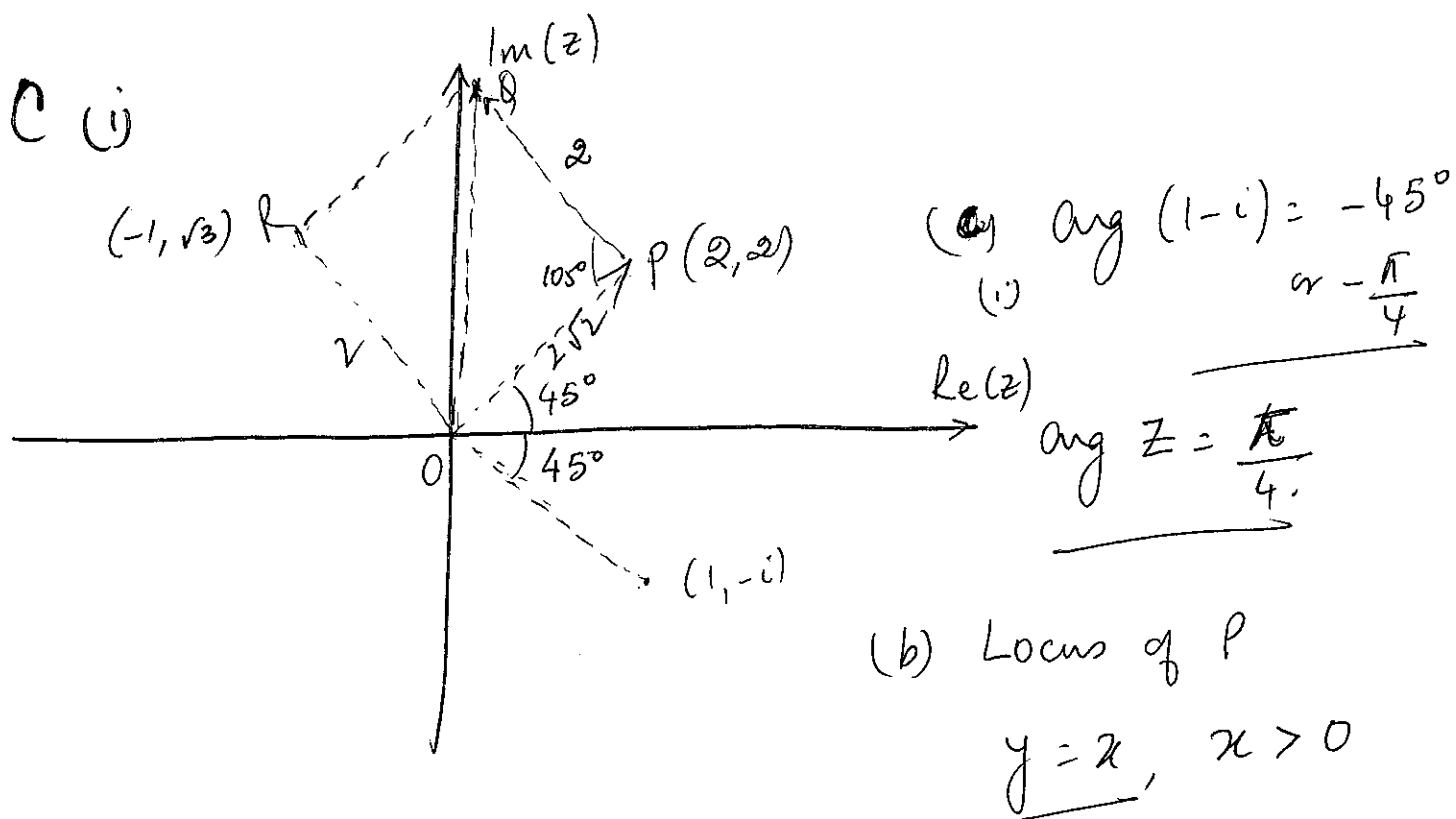
$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



(ii) $w = r(\cos\theta + i\sin\theta) \quad \bar{w} = r(\cos\theta - i\sin\theta)$
 $r^3(\cos 3\theta + i\sin 3\theta) = r^3(\cos 3\theta - i\sin 3\theta)$

$$2i\sin 3\theta = 0$$

$$3\theta = 0, \pi, 2\pi, 3\pi, \dots$$

$$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \dots$$

$$\arg(w) = \frac{2\pi}{3} \quad (\text{since } \frac{\pi}{2} < \theta < \pi)$$

(iii) $P \nparallel : 2\sqrt{2}(\cos 45^\circ + i\sin 45^\circ)$
 $= 2\sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)$
 $= 2 + 2i$

$R : 2\left(\cos \frac{2\pi}{3} + i\sin \frac{2\pi}{3}\right)$
 $= 2\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
 $= -1 + \sqrt{3}i$

$$\therefore Q = (2-1) + i(2+\sqrt{3})$$

$$\text{or } \underline{\underline{Q = 1 + (\sqrt{3}+2)i}}$$

3(b)

$$(i) \quad OQ : x b \cos \theta + y a \sin \theta = 0$$

$$y = -\frac{x b \cos \theta}{a \sin \theta}$$

$$(ii) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{Sub: } y = -\frac{x b \cos \theta}{a \sin \theta}$$

$$\frac{x^2}{a^2} + \frac{x^2 b^2 \cos^2 \theta}{a^2 \sin^2 \theta \cdot b^2} = 1$$

$$x^2 (1 + \cos^2 \theta) = a^2 \sin^2 \theta$$

$$x^2 \sin^2 \theta + x^2 \cos^2 \theta = a^2 \sin^2 \theta$$

$$\therefore x^2 = a^2 \sin^2 \theta$$

$$\therefore x = +a \sin \theta \quad (\because x > 0 \text{ in 1st quadrant})$$

$$y = -\frac{a b \sin \theta \cos \theta}{a \sin \theta}$$

$$= -b \cos \theta$$

$$\therefore Q (a \sin \theta, -b \cos \theta)$$

(iii) Perpendicular distance from $P(a \cos \theta, b \sin \theta)$ to OQ ($x b \cos \theta + y a \sin \theta = 0$) is

$$d = \left| \frac{ab \cos^2 \theta + ab \sin^2 \theta}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right| = \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$\text{Distance } OQ = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$\therefore \text{Area } \Delta OPQ = \frac{1}{2} OQ \times d$
which is independent of θ

Q4

$$(a) \quad P(x) = ax^3 + bx^2 + cx + d$$

$$d = -15 \quad \therefore P(x) = ax^3 + bx^2 + cx - 15$$

$$P(2) = 8a + 4b + 2c - 15 = 5$$

$$\text{or } 8a + 4b + 2c - 20 = 0$$

$$\text{or } 4a + 2b + c - 10 = 0 \quad \text{--- (1)}$$

one zero is $1 + 2i$

$\therefore$ another zero is $1 - 2i$ ($\because P(x)$ has real coefficients)

$$\alpha\beta\gamma = -(-15)/a$$

$$= 15/a$$

$$(1+2i)(1-2i)\gamma = \frac{15}{a}$$

$$5\gamma = \frac{15}{a}$$

$$\therefore \gamma = \frac{3}{a}$$

$$\therefore \cancel{P(3) = 0}$$

$$\alpha + \beta + \gamma$$

$$= 1 + 2i + 1 - 2i + \frac{3}{a} = -\frac{b}{a}$$

$$+ \frac{15}{a} \quad \frac{15}{a} \quad 2a + 3 = -b$$

$$2a + b + 3 = 0$$

$$+ \frac{15}{a} \quad \frac{15}{a} \quad 5a + b + 3 = 0 \quad \text{--- (2)}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha$$

$$(1+2i)(1-2i) + (1+2i)\frac{3}{a} + \frac{3}{a}(1+2i) = \frac{c}{a}$$

$$1 + 4 + \frac{3}{a} - \frac{6i}{a} + \frac{3}{a} + \frac{6i}{a} = \frac{c}{a}$$

$$11 + \frac{6}{a} = \frac{c}{a} \quad 5a + 6 = c$$

$$11a + 6 = c \quad 5a - c + 6 = 0$$

$$4a + 2b + c = 10 \quad \text{--- (1)}$$

$$2a + b = -3 \quad \text{--- (2)}$$

$$5a - c = -6 \quad \text{--- (3)}$$

$$\textcircled{1} + \textcircled{3}$$

$$9a + 2b = 4 \quad \text{--- (4)}$$

$$\textcircled{4} - 2 \times \textcircled{2}$$

$$5a = 10$$

$$\therefore a = 2$$

$$\therefore b = -3 - 4 = -7$$

$$c = -6 - 10 = -16$$

$$\therefore P(x) = 2x^3 - 7x^2 - 16x - 15 = 0$$

$$4(b) \text{ (i) } x^3 - 4x + 5 = 0 \quad \alpha + \beta + \gamma = 0, \quad \alpha\beta + \beta\gamma + \gamma\alpha = -4$$

$$\alpha\beta\gamma = -5$$

$$P(\alpha) = \alpha^3 - 4\alpha + 5 = 0 \quad \text{--- (1)}$$

$$P(\beta) = \beta^3 - 4\beta + 5 = 0 \quad \text{--- (2)}$$

$$P(\gamma) = \gamma^3 - 4\gamma + 5 = 0 \quad \text{--- (3)}$$

$$\text{(1) + (2) + (3)} \quad \alpha^3 + \beta^3 + \gamma^3 - 4(\alpha + \beta + \gamma) + 15 = 0$$

$$\underline{\alpha^3 + \beta^3 + \gamma^3 = -15}$$

$$\text{(ii) } \alpha + \beta = (\alpha + \beta + \gamma) - \gamma$$

$$= 0 - \gamma = -\gamma$$

$$\beta + \gamma = -\alpha$$

$$\gamma + \alpha = -\beta$$

$\therefore$ The roots of the new equation are $-\alpha, -\beta, -\gamma$

$$\text{let } y = -x$$

$$P(x) = x^3 - 4x + 5 = 0$$

$$P(-y) = (-y)^3 - 4(-y) + 5 = 0$$

$$-y^3 + 4y + 5 = 0$$

$$\text{or } \underline{x^3 - 4x - 5 = 0}$$

$$\therefore \text{(iii) } (\alpha + \beta)^2 (\beta + \gamma)^2 (\gamma + \alpha)^2 = [\alpha\beta(\beta\gamma)(\gamma\alpha)]^2$$

$$= (\text{pde of roots})^2$$

$$= [-(-5)]^2 = \boxed{25}$$

(c) Factorise $x^4 - 2x^2 - 15$ ~~450~~

(i) $(x^2 - 5)(x^2 + 3)$ ~~450~~

(ii) $(x + \sqrt{5})(x - \sqrt{5})(x + \sqrt{3}i)(x - \sqrt{3}i)$ ~~450~~